

```

• import sys
• from collections import defaultdict
• from io import StringIO
• from matplotlib import pyplot as plt
• from PIL import Image
• sys.path.append(".")
• From object_detection.utils import ops as utils_ops
• From utils import label_map_util
• from utils import visualization_utils as vis_util

```

Just a small description of what this does. On executing the commands shown previously, TensorFlow will be updated with the numpy library for numerical tasks, matplotlib for graphical representations and zipfile for extracting models from the github repository that are saved in compressed files.

After this, add and import the libraries for neural networks[NN], feed-back loops and feed forward networks. Object detection requires the models and libraries for image conversion, re-sampling and reassembling which differ from system to system.

To upgrade all the download extensions and files when using the environment again once it's been shut off, initialise all the softwares through the conda terminal and type:-

```

• pip install --upgrade tensorflow

```

These steps will thus prepare you for executing batch files, compiling the code and working on object detection programs. You can also use the following command to update TensorFlow when working as an administrator in the command prompt terminal:-

```
C:\> pip3 install --upgrade tensorflow
```

TESTING THE ENVIRONMENT

Make a simple program in the terminal other than the hello world command, say perhaps for finding the squares and plotting them against their respective roots. The purpose to this is to test the functionality of the pip commands and the libraries.

```

• int i=0;
• int x[10];

```